



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to ANDAVAR AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

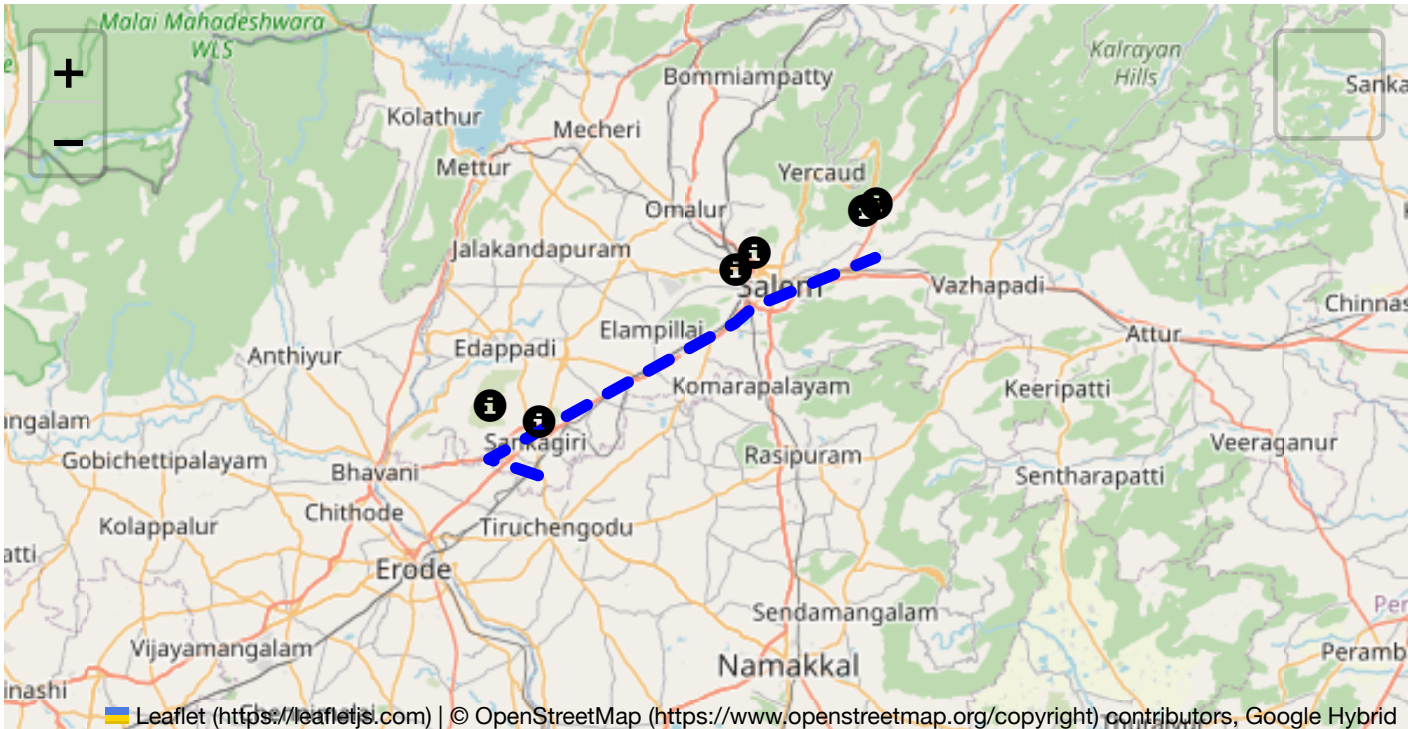
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

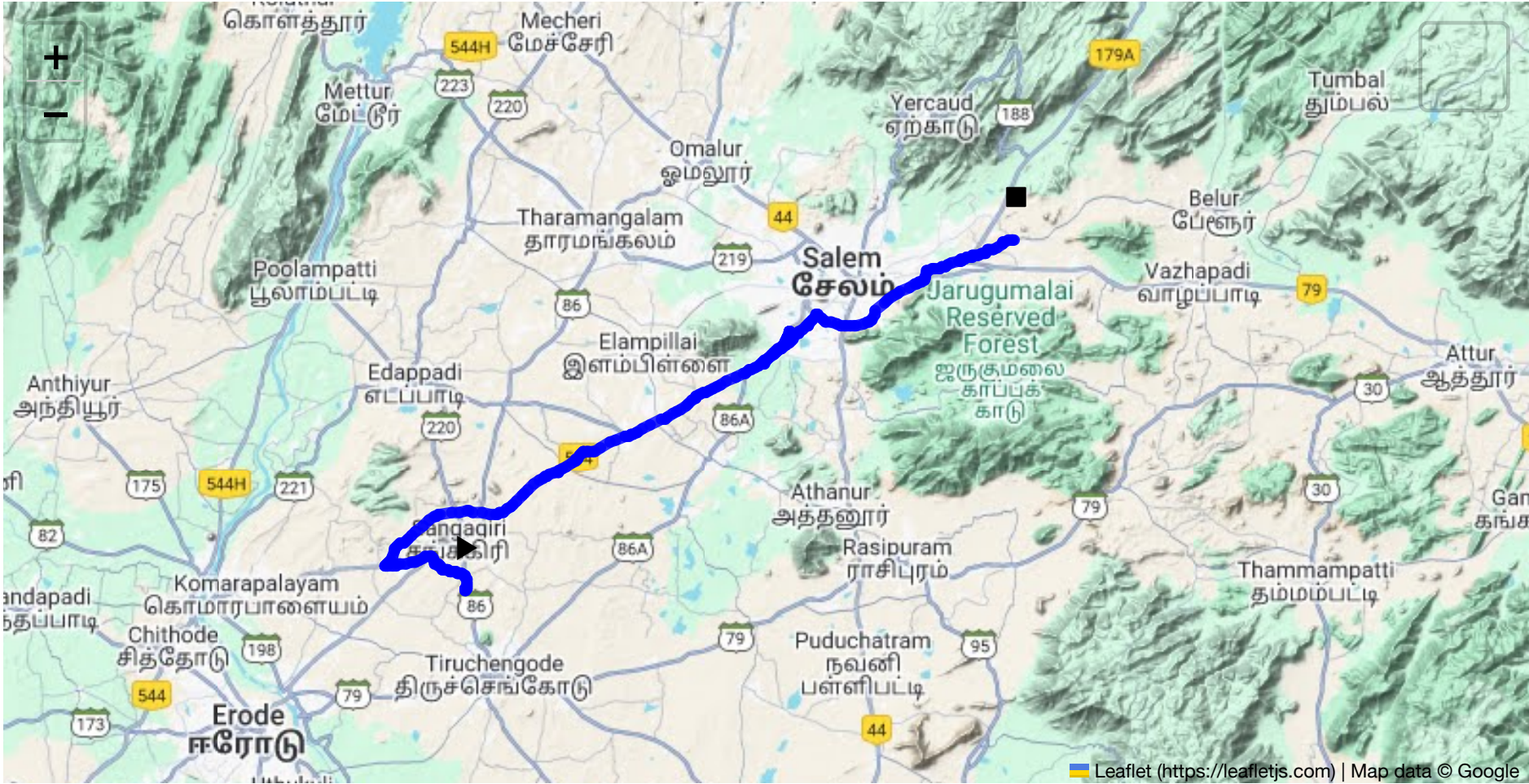
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 76.25 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours

Start: (11.4381, 77.8734)
End: (11.686043, 78.26918)



Welcome to the Journey Risk Management Study

Route Map Overview: The route spans approximately 76.25 kilometers from Sangagiri to Kullampatti, passing through key points in Tamil Nadu including Bhavani Main Road, Nagapattinam - Coimbatore - Gundlupet Hwy, and Ayothiyapattinam - Belur - Thumbal Main Rd. It is primarily a mix of highways and smaller state roads that traverse both urban and rural areas.

Typical Weather Conditions and Potential Hazards:

- Tamil Nadu generally experiences tropical weather, with hot summers, and monsoon season typically between June and September.
- Heavy rains during the monsoon can lead to waterlogging and reduced visibility, increasing the risk of accidents.
- Occasional fog in the early morning hours during the winter months (December to February) may also affect visibility.

Traffic Patterns and Congestion Areas:

- Traffic congestion is common in urban centers like Salem, particularly during peak hours (8:00-10:00 AM and 5:00-7:00 PM).
- Bhavani Main Road and areas near Kondalampatty experience significant traffic due to market activities and commuters.
- It's advisable to plan the journey to avoid peak times if possible.

Road Quality and Infrastructure:

- The Nagapattinam - Coimbatore - Gundlupet Hwy is generally well-maintained, with good road conditions suitable for heavy vehicles.
- Smaller roads, like those connecting Sangagiri to Padaiveedu, might have uneven patches and require careful navigation.
- Road signs and markings may be insufficient in rural parts, necessitating extra caution.

Alternative Routes for Emergencies:

- In case of blockages or emergencies, divert to National Highway 44 for a consistent route with better facilities.
- Smaller state highways can also serve as detours, but verify their condition during the monsoon season before choosing them.

Local Regulations for Hazardous Material Transport:

- Ensure compliance with Tamil Nadu's regulations on hazardous materials, which include restrictions on transport times and mandated rest stops.
- Permits and clear documentation are required, and regular checks are enforced at certain checkpoints.

Historical Incidents Analysis:

- No major reports of incidents specific to this route involving heavy vehicles or hazardous materials.
- However, general accident rates along busy highway sections necessitate vigilance.

Environmental Considerations and Sensitive Areas:

- Several towns along the route may have residential zones requiring reduced speed.
- Avoidance of ecologically sensitive areas such as natural reserves will be required by regulations where applicable.

Communication Coverage:

- Major parts of the route, including highways and urban centers, have good mobile network coverage.
- Potential dead zones may exist in rural stretches. Carrying an alternative communication device is advisable.

Emergency Response Times:

- Emergency response in urban areas such as Salem is relatively quick, estimated at 15-30 minutes.
- Response in rural sections like Padaiveedu or Sangagiri can take longer, potentially up to an hour depending on the exact location.

Overall Risk Assessment:

- The route generally poses moderate risk for transporting hazardous materials, given the mix of highway and rural road infrastructure.
- Key risks include monsoon-related weather challenges, congestion in urban areas, and occasional coverage issues in rural parts.
- Proactive planning regarding timing, weather conditions, and communication readiness significantly mitigates these risks.

In summary, effective journey management, adherence to regulations, and real-time traffic and weather updates are crucial for safely navigating this route with heavy vehicles transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	U-Turn	High	11.5991031, 78.0887685	10 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45358, 77.85781	30 KM/Hr
7	Turn	High	11.45353, 77.85697	15 KM/Hr
8	Turn	Medium	11.45853, 77.85139	30 KM/Hr
9	Turn	Medium	11.45906, 77.85132	30 KM/Hr
10	Turn	Medium	11.46307, 77.84941	30 KM/Hr
11	Turn	Medium	11.46316, 77.84921	30 KM/Hr
12	Turn	High	11.45542, 77.81725	15 KM/Hr
13	Turn	High	11.45636, 77.81678	15 KM/Hr
14	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
15	Turn	High	11.62130, 78.10776	15 KM/Hr
16	Turn	Medium	11.61205, 78.10427	30 KM/Hr
17	Turn	High	11.59910, 78.08877	15 KM/Hr
18	Blind Spot	Blind Spot	11.59918, 78.08868	10 KM/Hr
19	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
20	Turn	High	11.62130, 78.10776	15 KM/Hr
21	Blind Spot	Blind Spot	11.61217, 78.10439	10 KM/Hr
22	Turn	Medium	11.62498, 78.12011	30 KM/Hr
23	Turn	High	11.63297, 78.12749	15 KM/Hr
24	Turn	Medium	11.62826, 78.13337	30 KM/Hr
25	Turn	Medium	11.65632, 78.20220	30 KM/Hr
26	Turn	High	11.66380, 78.20648	15 KM/Hr
27	Turn	High	11.67120, 78.23136	15 KM/Hr
28	Turn	Medium	11.67190, 78.23144	30 KM/Hr
29	Turn	Medium	11.67672, 78.24922	30 KM/Hr
30	Turn	Medium	11.67944, 78.25198	30 KM/Hr
31	Turn	Medium	11.68547, 78.26757	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
22	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
23	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
25	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
28	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium
29	school	Holy Cross Matriculation Higher Secondary School	11.6635657, 78.203532	30 km/h	Medium
30	school	Govt Hr Sec School	11.6754733, 78.2459724	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.5991031, 78.0887685



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61205, 78.10427



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59910, 78.08877



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.59918, 78.08868



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61217, 78.10439



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62498, 78.12011



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63297, 78.12749



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62826, 78.13337



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65632, 78.20220



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66380, 78.20648



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67120, 78.23136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67190, 78.23144



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67672, 78.24922



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67944, 78.25198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68547, 78.26757

